

Name Key

Physics 390 In-Class Activity 11

(DO NOT INCLUDE SOURCES WHEN YOU BACK PROJECT)

OBJECT						RECONSTRUCTION					
					P1						P1
0	0	0	0	0	∅	∅	1	14	1	∅	∅
0	0	0	0	0	∅	∅	1	14	1	∅	∅
0	0	12	0	0	12	12	13	26	13	12	12
0	1	2	1	0	4	4	5	18	5	4	4
0	0	0	0	0	∅	∅	1	14	1	∅	∅
∅	1	14	1	∅	P2	∅	1	14	1	∅	P2

Red# = Blue# + Green#

Questions

1. The bulk of the attenuating material is in the center of the object. Is this also true for the image produced by this very rudimentary simple back projection?

Yes - the bulk of the attenuating material is in the center of the reconstruction

2. Does this back projection suffer from blur (i.e., attenuating material appearing in incorrect parts of the image)?

Yes

3. Do you think that medicine can do better?

Yes